

HW7

Devon Park

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Question 1:

Fit a Poisson regression model with hospital length of stay as the outcome and procedure, gender, age, and type as covariates (model 1). Write down the model. Is overdispersion a potential problem for this Poisson model?

Table 1: Poisson Model (Model 1)

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.555	0.150	3.695	0	0.260	0.849
procedure_grpCABG	1.122	0.019	59.137	0	1.085	1.159
gender_grpMale	-0.103	0.018	-5.654	0	-0.138	-0.067
age	0.010	0.002	4.913	0	0.006	0.014
type_grpEmergency/Urgent	0.190	0.017	11.275	0	0.157	0.223

General Model Form: $\log E(Y \mid \text{procedure, gender, age, type}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$

Y = length of hospital stay in days x_1 = procedure where 1 = CABG, 0 = PTCA x_2 = gender where 1 = male, 0 = female x_3 = age in years x_4 = type where 1 = emergency/urgent, 0 = elective

Model with Coefficients: General Model Form: $\log E(Y) = 0.555 + 1.122x_1 - 0.103x_2 + 0.010x_3 + 0.190x_4$

To check overdispersion we can use quasipoisson method:

Table 2: Quasipoisson Model

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.555	0.224	2.479	0.013	0.116	0.993
procedure_grpCABG	1.122	0.028	39.671	0.000	1.067	1.178
gender_grpMale	-0.103	0.027	-3.793	0.000	-0.156	-0.050
age	0.010	0.003	3.296	0.001	0.004	0.016
type_grpEmergency/Urgent	0.190	0.025	7.564	0.000	0.141	0.239

The dispersion parameter is calculated as 2.22. As this value is greater than 1, this indicates overdispersion where $\text{Var}(Y) \sim 2.22 \cdot E(Y)$. As Poisson assumes that variance and mean are equal, this parameter suggests that the actual variance is more than 2x larger than what model 1 assumes (and therefore the standard errors of model 1 are too small). We should address overdispersion.

Question 2:

Refit model 1 with the scale parameter being equal to Pearson chi-square divided by residual DF. Estimate the length of stay rate ratio between patients undergoing CABG and PTCA procedures. Provide the 95% confidence interval and interpret When I ran `quasi_model` above, it already recalculated using the scale parameter being equal to `Pearson chi-square/ DF`. (Estimates stay the same and the standard errors are recalculated).

Table 3: Quasipoisson Rate Ratios

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1.741	0.224	2.479	0.013	1.123	2.700
procedure_grpCABG	3.071	0.028	39.671	0.000	2.906	3.246
gender_grpMale	0.902	0.027	-3.793	0.000	0.856	0.952
age	1.010	0.003	3.296	0.001	1.004	1.016
type_grpEmergency/Urgent	1.209	0.025	7.564	0.000	1.151	1.270

To estimate the los rate ratio between patients undergoing CABG and PTCA procedures, we look at the procedures covariate. We get the following values: $\text{beta}(\text{procedure}) = 1.122$ $\text{RateRatio} = 3.071$ 95

Interpretation: The estimated los for patients undergoing CABG is 3.071 times that of patients undergoing the PTCA procedure after adjusting for age, gender, and admission type. We are 95% confident that the true risk ratio exists between 2.906 and 3.246. At $\alpha = 0.05$, these values are statistically significant as the p-value of the coefficient is «0.001.

Question 3:

Use the fitted model in part (2), calculate the expected days of hospital stay for male patients aged 68 who underwent CABG procedure and stayed in an elective type My attempt to do this via code:

```
##          1
## 9.68651
```

The output is 9.69 days.

We can also do this manually by plugging into our model 1.

$$\log E(Y) = 0.555 + 1.122(1) - 0.103(1) + 0.010(68) + 0.190(0) \quad \log E(Y) = 0.555 + 1.122 - 0.103 + 0.680 + 0 = 2.254$$

$$\exp(2.254) = 9.687 \text{ days}$$

Woohoo I get the same results!

Interpretation: A 68-year-old male patient who has an elective CABG procedure is expected to stay in the hospital for approximately 9.69 days.

Question 4:

Refit model 1 using negative binomial regression. Provide a formal test to decide whether a negative binomial model is needed for this data than a Poisson regression model.

Table 4: Negative Binomial Model

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.543	0.208	2.614	0.009	0.137	0.948
procedure_grpCABG	1.131	0.024	46.633	0.000	1.083	1.178
gender_grpMale	-0.107	0.025	-4.242	0.000	-0.157	-0.058

term	estimate	std.error	statistic	p.value	conf.low	conf.high
age	0.010	0.003	3.529	0.000	0.005	0.016
type_grpEmergency/Urgent	0.218	0.023	9.328	0.000	0.172	0.264

From `nb_model`, theta ($1/\alpha$) is 9.292 with a standard error of 0.668.

H0: alpha equals zero and poisson model is good HA: alpha does not equal zero, so negative binomial model is needed Test statistic: $z = \theta / SE(\theta) = 9.292 / 0.668 = 13.91$ When $z = 13.91$, $p\text{-value} < 0.001$.

Interpretation: at $\alpha = 0.05$, we reject the null hypothesis. There is statistically significant evidence that the dispersion parameter is different from zero. This means that the negative binomial model is more appropriate than the Poisson model for this data.

Question 5:

Use the negative binomial model to estimate the length of stay rate ratio between patients undergoing CABG and PTCA procedures and provide 95% confidence interval. Is the conclusion different from the Poisson model in part (2)?

Table 5: Negative Binomial Rate Ratios

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1.721	0.208	2.614	0.009	1.147	2.580
procedure_grpCABG	3.098	0.024	46.633	0.000	2.954	3.249
gender_grpMale	0.898	0.025	-4.242	0.000	0.855	0.944
age	1.010	0.003	3.529	0.000	1.005	1.016
type_grpEmergency/Urgent	1.244	0.023	9.328	0.000	1.188	1.302

From the earlier code for `nb_model`, the estimate for procedure ($\beta(\text{procedure})$) is 1.131. The rate ratio between patients undergoing CABG and PTCA procedures is 3.098. The 95% confidence interval is (2.95, 3.249).

Interpretation: Based on the negative binomial model, the estimated length of hospital stay for patients undergoing CABG is 3.098 times that of patients undergoing the PTCA procedure, after adjusting for gender, age, and admission type. We are 95% confident that the true rate ratio lies between 2.954 and 3.249. This result is statistically significant at the pre-specified alpha of 0.05 as the p-value of the estimator is < 0.001 .

After comparing both the poisson model and the negative binomial model we see that the estimated rate ratios and their confidence intervals are extremely similar. So, while our conclusion doesnt change significantly after using negative binomial it is still the more appropriate model because we found overdispersion earlier.